

UAE Post-Quantum Cryptography Roadmap

OG RMM Platform — Cryptographic Inventory and Migration Plan

Document Reference: OG-RMM-PQC-001

Version: 1.0

Date: March 2026

Classification: Confidential

Applicable Standards: UAE Cybersecurity Council PQC Directive (2025), NIST FIPS 203/204/205, ETSI TR 103 619

Prepared by: Manus AI — Security Architecture

1. Executive Summary

The UAE Cybersecurity Council issued a Post-Quantum Cryptography (PQC) preparedness directive in 2025, requiring all critical infrastructure operators — including oil and gas SCADA systems — to complete a cryptographic inventory and submit a migration roadmap by Q4 2026. This document fulfils that requirement for the OG RMM Platform, cataloguing all cryptographic primitives in use, assessing quantum vulnerability, and providing a phased migration plan to NIST-standardised post-quantum algorithms (FIPS 203 ML-KEM, FIPS 204 ML-DSA, FIPS 205 SLH-DSA).

The platform's current cryptographic posture is **Amber** — all classical algorithms in use are quantum-vulnerable over a 10–15 year horizon, but no immediate threat exists. The migration plan targets **Green** status by Q2 2028 through a hybrid classical/PQC approach that maintains backward compatibility throughout the transition.

2. Cryptographic Inventory

2.1 Transport Layer Cryptography

Component	Protocol	Algorithm	Key Size	Quantum Vulnerability	Migration Priority
API Gateway ↔ Frontend	TLS 1.3	X25519 + AES-256-GCM	256-bit	High (ECDH)	P1 — Hybrid X25519Kyber768
Edge Agent ↔ Stream Processor	mTLS 1.3	X25519 + AES-256-GCM	256-bit	High (ECDH)	P1 — Hybrid X25519Kyber768
API Gateway ↔ Microservices	TLS 1.3	X25519 + AES-256-GCM	256-bit	High (ECDH)	P2 — Internal migration
Redpanda inter-broker	TLS 1.3	X25519 + AES-256-GCM	256-bit	High (ECDH)	P2
PostgreSQL client ↔ server	TLS 1.3	X25519 + AES-256-GCM	256-bit	High (ECDH)	P2
MQTT (Mosquitto)	TLS 1.3	X25519 + AES-256-GCM	256-bit	High (ECDH)	P1 — Field device exposure

2.2 Authentication and Digital Signatures

Component	Algorithm	Key Size	Quantum Vulnerability	Migration Priority
JWT tokens (API Gateway)	HMAC-SHA256	256-bit	Low (symmetric)	P3 — Increase to SHA-512
JWT tokens (refresh)	RSA-4096	4096-bit	High (RSA)	P1 — Migrate to ML-DSA-65
Keycloak SAML assertions	RSA-2048	2048-bit	High (RSA)	P1 — Migrate to ML-DSA-44
Temporal workflow signatures	ECDSA P-256	256-bit	High (ECDSA)	P1 — Migrate to ML-DSA-44
TLS certificates (CA)	RSA-4096	4096-bit	High (RSA)	P1 — Migrate to ML-DSA-87
TLS certificates (leaf)	ECDSA P-256	256-bit	High (ECDSA)	P1 — Migrate to ML-DSA-44
Code signing (Docker images)	ECDSA P-256	256-bit	High (ECDSA)	P2 — Migrate to SLH-DSA-128s
Git commit signatures	ECDSA P-256	256-bit	High (ECDSA)	P3

2.3 Data-at-Rest Encryption

Component	Algorithm	Key Size	Quantum Vulnerability	Migration Priority
PostgreSQL tablespace encryption	AES-256-CBC	256-bit	Low (symmetric)	P3 — Migrate to AES-256-GCM
MinIO/S3 server-side encryption	AES-256-GCM	256-bit	Low (symmetric)	None required
Redis AOF encryption	AES-256-GCM	256-bit	Low (symmetric)	None required
InfluxDB data files	AES-256-GCM	256-bit	Low (symmetric)	None required
Kubernetes Secrets (etcd)	AES-256-GCM	256-bit	Low (symmetric)	None required
Key Management (Vault)	AES-256-GCM + RSA-4096	256/4096-bit	High (RSA wrapping)	P1 — Migrate to ML-KEM-768

2.4 Key Exchange / Key Encapsulation

Component	Algorithm	Quantum Vulnerability	Migration Target
TLS key exchange	X25519 ECDH	High	ML-KEM-768 (FIPS 203) hybrid
Vault key wrapping	RSA-OAEP-4096	High	ML-KEM-1024 (FIPS 203)
MQTT session keys	X25519 ECDH	High	ML-KEM-768 hybrid
Keycloak token encryption	RSA-OAEP-2048	High	ML-KEM-768

3. Risk Assessment

3.1 Harvest-Now-Decrypt-Later (HNDL) Risk

The most immediate quantum threat to the OG RMM Platform is the HNDL attack vector, where adversaries record encrypted OT traffic today with the intention of decrypting it once cryptographically relevant quantum computers (CRQCs) become available. Given that:

- NIST estimates CRQCs capable of breaking RSA-2048 and ECDH-256 will emerge between 2030 and 2035.
- The OG RMM Platform handles production data, financial transactions, and safety system configurations with a 10–30 year operational lifespan.
- Kuwait and UAE are geopolitically significant energy producers with state-level adversary interest.

The HNDL risk is classified as **High** for financial ledger data and **Medium** for operational telemetry.

3.2 Algorithm Vulnerability Timeline

Algorithm	Classical Security	Quantum Security	Estimated Break Year
RSA-2048	112-bit	0-bit (Shor's)	2030–2035
RSA-4096	140-bit	0-bit (Shor's)	2033–2038
ECDSA P-256	128-bit	0-bit (Shor's)	2030–2035
X25519 ECDH	128-bit	0-bit (Shor's)	2030–2035
AES-256-GCM	256-bit	128-bit (Grover's)	Never (sufficient)
HMAC-SHA256	256-bit	128-bit (Grover's)	Never (sufficient)
SHA-256	256-bit	128-bit (Grover's)	Never (sufficient)

4. Migration Plan

Phase 1: Hybrid PQC Deployment (Q3 2026 – Q1 2027)

The hybrid approach deploys PQC algorithms alongside classical algorithms, ensuring backward compatibility with all existing clients while providing quantum resistance for new connections.

P1 Actions:

Action	Algorithm	Component	Effort	Owner
Deploy hybrid TLS (X25519Kyber768) on API Gateway	ML-KEM-768 + X25519	Go API Gateway	3 weeks	Security Eng.
Deploy hybrid TLS on edge agent mTLS	ML-KEM-768 + X25519	Rust Edge Agent	2 weeks	Security Eng.
Migrate JWT refresh tokens to ML-DSA-65	ML-DSA-65 (FIPS 204)	Go API Gateway	2 weeks	Security Eng.
Migrate Keycloak SAML to ML-DSA-44	ML-DSA-44 (FIPS 204)	Keycloak	1 week	IAM Eng.
Migrate Vault key wrapping to ML-KEM-1024	ML-KEM-1024 (FIPS 203)	HashiCorp Vault	2 weeks	Infra Eng.
Migrate TLS certificates to ML-DSA-87	ML-DSA-87 (FIPS 204)	PKI / cert-manager	4 weeks	Infra Eng.

Phase 2: Internal Service Migration (Q2 2027 – Q4 2027)

P2 Actions:

Action	Algorithm	Component	Effort
Migrate inter-service TLS to hybrid PQC	ML-KEM-768 + X25519	All Go/Rust services	4 weeks
Migrate Redpanda broker TLS	ML-KEM-768 + X25519	Redpanda	1 week
Migrate PostgreSQL client TLS	ML-KEM-768 + X25519	All DB clients	2 weeks
Migrate Docker image signing to SLH-DSA	SLH-DSA-128s (FIPS 205)	CI/CD pipeline	2 weeks

Phase 3: Legacy Removal and Full PQC (Q1 2028 – Q2 2028)

P3 Actions:

Action	Component	Effort
Remove classical fallback from hybrid TLS	API Gateway, Edge Agent	1 week
Upgrade JWT HMAC-SHA256 to HMAC-SHA512	API Gateway	1 week
Migrate Git commit signing to SLH-DSA	Developer workstations	2 weeks
Final cryptographic audit and UAE Cybersecurity Council submission	All	3 weeks

5. NIST PQC Algorithm Reference

NIST Standard	Algorithm	Type	Security Level	Use Case
FIPS 203	ML-KEM (Kyber)	Key Encapsulation	128/192/256-bit	Key exchange, TLS
FIPS 204	ML-DSA (Dilithium)	Digital Signature	128/192/256-bit	Authentication, JWT, TLS certs
FIPS 205	SLH-DSA (SPHINCS+)	Digital Signature	128/192/256-bit	Code signing, long-lived certs

6. UAE Cybersecurity Council Submission Checklist

Requirement	Status	Evidence
Cryptographic inventory complete	Complete	Section 2 of this document
Quantum vulnerability assessment	Complete	Section 3 of this document
Migration roadmap with timeline	Complete	Section 4 of this document
NIST FIPS 203/204/205 adoption plan	Complete	Section 4, Phase 1–3
Hybrid PQC deployment timeline	Q3 2026	Phase 1 plan
Full PQC deployment timeline	Q2 2028	Phase 3 plan
Responsible officer signature	Pending	CISO sign-off required

This document must be reviewed by the CISO and submitted to the UAE Cybersecurity Council by Q4 2026 as required by the PQC Preparedness Directive.